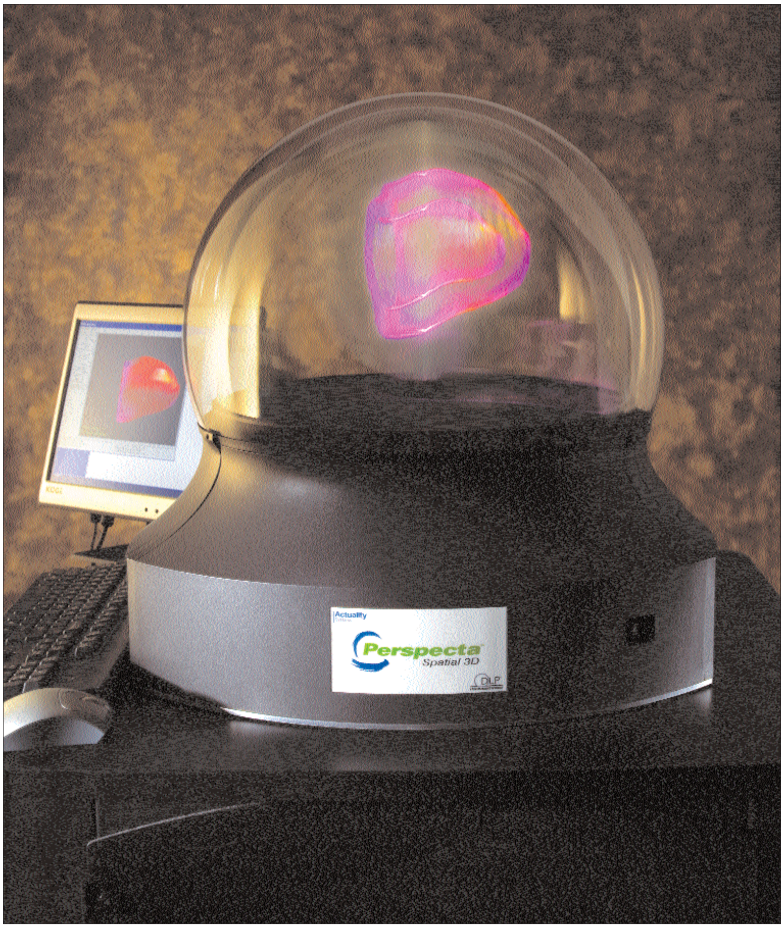

The Perspecta Spatial 3D System's 20-inch dome, displaying a 3D image of a heart. It occupies a volume in space, giving users an all-encompassing view
Image courtesy of Actuality Systems Inc., Bedfort, MA, USA © 2004 David Shoppen

This, however, was not Garner's biggest challenge: remember that for any 3D image to materialise, a volume that acts as a screen is required. That was the challenge—to find a suitable projection surface that has a volume.

A column of mist should be OK, because it has particulates on which light can be reflected. But mist diffuses the projected image. Garner is therefore working with a display composed of layers of micro-thin LCD panels, each of which can (when electrically charged) be made clear or opaque. The panels flash on and off in quick such quick succession while assemble the hologram that the speed convinces the eye it's seeing a solid object.

Other such displays exist today, but instead of using a series of 2D interferograms, they slice up a 3D image and send the 2D data sliver by sliver to the LCD screens. This method requires far greater processing power. Which is why Garner's approach is the most viable solution for 3D TV. "We're sending the 3D images as a 2D interferogram," Garner says, and this doesn't require any more bandwidth than today's TV signals—"so we can use the current broadcast infrastructure."

As for creating the holographic content, it would have to be recorded with a series of cameras shooting from different viewpoints.

Dr Garner and his research team—Dr Huebschman and programmer Bala Munjuluri—have published details of their system in several publications. For technical details and sample holographic movies, see **http://innovation.swmed.edu/research/instrumentation/res_inst_dev3d.html**. What we've provided here is a rough overview, and to understand the system in detail, a good knowledge of interferograms is essential.

Think of Garner's assembly as Dr Bell talking on his 'telephone' with his assistant Watson, compared with our casual usage of cell phones today. 2D interferograms bounced off the million mirrors of a DMD may well be the prototype that leads to 3D TV—who knows?

An Enclosed Volumetric Display

Perspecta Spatial 3D (www.actuality-systems.com) urges, "End Flat-Screen Thinking."

Think of it as a plug-and-play crystal ball. The Perspecta Spatial 3D System includes a 20-inch dome displaying full-colour and motion images that occupy a volume in space, giving users a 360 degree all-encompassing view-and without the need for goggles.

Perspecta renders all movement from widely-used open-standard 3D applications. The resolution of the gizmo is a 100 million voxels! A voxel is a volumetric pixel.

The Web site proclaims, "Gone are the days when colleagues squeezed in behind your desk to view an HIV molecule or patient MRI. With Perspecta Spatial 3D, everyone sees everything all the time." The system exploits the fact that the brain can integrate a series of 2D cross-section images (the device spins at 730 rpm) into a volume-filling 3D image.

From a whitepaper on the site, we learnt that spatial 3D is different from flat-screen 3D projection or the use of stereoscopic goggles. Spatial 3D refers to 3D imagery that truly occupies a volume of space. The scientific term for a spatial display is a 'volumetric display'.

The Perspecta Spatial 3D System projects 3D imagery that seems to hover inside a transparent dome. "Spatial 3D gives you a unique perspective on information because you can walk around the imagery to inspect it from different angles." And the site emphasises that the 3D imagery is "truly three-dimensional; it isn't an illusion and doesn't require wearing any goggles."

The Perspecta system is probably the closest we've gotten so far to the holy grail of true 3D projection. So what's missing? Not too much, except that the 'crystal ball' isn't too large, and that you can't interact with it in any way. And, of course, it's expensive—$40,000 (more than Rs 17 lakh). But still, the system does blur the line between sci-fi and actuality—pun intended!

The Way Ahead

The list of systems we've mentioned above is in no way exhaustive. There are virtually tons of different 3D visualisation systems you'll find if you Google the appropriate terms. We haven't bothered with mentioning more of them here because there are just too many!

One thing we can expect is for such systems to become more common. For example, in July